

Часть 1

8.1.29

$$\int \frac{dx}{x^{\frac{5}{2}} \sqrt{x}} = \int \frac{dx}{x^{\frac{5}{2}} \cdot x^{\frac{1}{2}}} = \int \frac{dx}{x^{\frac{6}{2}}} = \int x^{-3} dx = \left[\int x^k dx = \frac{x^{k+1}}{k+1} + C \right] =$$

$$= \frac{x^{-\frac{6}{2}+1}}{-\frac{6}{2}+1} + C = \frac{x^{-\frac{3}{2}}}{-\frac{3}{2}} + C = -\frac{2}{3\sqrt{x^3}} + C = -\frac{2}{3x\sqrt{x}} + C$$

Проверка: $\left(-\frac{2}{3x\sqrt{x}} + C \right)' = \left(-\frac{2}{3} \cdot x^{-\frac{3}{2}} + C \right)' =$

$$= -\frac{2}{3} \cdot \left(-\frac{3}{2} \right) \cdot x^{-\frac{3}{2}-1} = x^{-\frac{5}{2}} = \frac{1}{x^{\frac{5}{2}} \sqrt{x}}$$

8.1.30

$$\int \frac{dx}{x^2+3} = \int \frac{dx}{x^2 + (\sqrt{3})^2} = \left[\int \frac{dx}{x^2+a^2} = \frac{1}{a} \operatorname{arctg} \frac{x}{a} + C \right] =$$

$$= \frac{1}{\sqrt{3}} \operatorname{arctg} \frac{x}{\sqrt{3}} + C = \frac{\sqrt{3}}{3} \operatorname{arctg} \frac{\sqrt{3}x}{3} + C$$

Проверка: $\left(\frac{\sqrt{3}}{3} \operatorname{arctg} \frac{\sqrt{3}x}{3} + C \right)' = \frac{\sqrt{3}}{3} \cdot \frac{1}{1 + \left(\frac{\sqrt{3}x}{3} \right)^2} \cdot \frac{\sqrt{3}}{3} =$

$$= \frac{1}{3} \cdot \frac{1}{\frac{3}{3} + \frac{x^2}{3}} = \frac{1}{3} \cdot \frac{3}{3+x^2} = \frac{1}{3+x^2}$$

8.1.31

$$\int \frac{1}{5^x} dx = \int \left(\frac{1}{5} \right)^x dx = \left[\int a^x dx = \frac{a^x}{\ln a} + C \right] = \frac{\left(\frac{1}{5} \right)^x}{\ln \frac{1}{5}} + C =$$

$$= \frac{\frac{1}{5^x}}{\ln 5^{-1}} + C = -\frac{1}{5^x \cdot \ln 5} + C$$

Проверка: $\left(-\frac{1}{5^x \ln 5} + C \right)' = -\frac{1}{\ln 5} \cdot \frac{1 \cdot (5^x)' - 5^x \cdot 1}{5^{2x}} =$

$$= -\frac{1}{\ln 5} \cdot \left(-\frac{\ln 5 \cdot 5^x}{5^{2x}} \right) = \frac{1}{5^x}$$

8.1.52

$$\int \frac{dx}{\sqrt{4-x^2}} = \int \frac{dx}{\sqrt{2^2-x^2}} = \left[\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin \frac{x}{a} + C \right] =$$

$$= \arcsin \frac{x}{2} + C.$$

Проверка: $\left(\arcsin \frac{x}{2} + C \right)'_x = \frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}} \cdot \frac{1}{2} = \frac{1}{2\sqrt{1-\frac{x^2}{4}}} =$

$$= \frac{1}{\sqrt{4-x^2}}.$$

8.1.53

$$\int \frac{dx}{\sqrt{x^2-1}} = \left[\int \frac{dx}{\sqrt{x^2+a^2}} = \ln|x+\sqrt{x^2+a^2}| + C \right] =$$

$$= \ln|x+\sqrt{x^2-1}| + C.$$

Проверка: $\left(\ln|x+\sqrt{x^2-1}| + C \right)'_x = \frac{1}{x+\sqrt{x^2-1}} \cdot \left(1 + \frac{2x}{2\sqrt{x^2-1}} \right) =$

$$= \frac{1}{x+\sqrt{x^2-1}} \cdot \frac{2(x+\sqrt{x^2-1})}{2\sqrt{x^2-1}} = \frac{1}{\sqrt{x^2-1}}.$$

8.1.34

$$\int \frac{dx}{x^2-25} = \int \frac{dx}{x^2-5^2} = \left[\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \cdot \ln \left| \frac{x-a}{x+a} \right| + C \right] =$$

$$= \frac{1}{10} \ln \left| \frac{x-5}{x+5} \right| + C.$$

Проверка: $\left(\frac{1}{10} \ln \left| \frac{x-5}{x+5} \right| + C \right)'_x = \frac{1}{10} \cdot \frac{x+5}{x-5} \cdot \left(\frac{(x+5)-(x-5)}{(x+5)^2} \right) =$

$$= \frac{1}{10} \cdot \frac{x+5}{x-5} \cdot \frac{10}{(x+5)^2} = \frac{1}{(x-5) \cdot (x+5)} = \frac{1}{x^2-25}.$$

$$\begin{aligned}
 8.1.35. \quad \int \left(x + \frac{2}{x}\right)^2 dx &= \int \left(x^2 + \frac{4x}{x} + \frac{4}{x^2}\right) dx = \\
 &= \left[\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx \right] = \int x^2 dx + \int \frac{4x}{x} dx + \\
 &+ \int \frac{4}{x^2} dx = \int x^2 dx + \int 4 dx + 4 \int \frac{dx}{x^2} = \left[\begin{array}{l} 1. \int x^2 dx = \frac{x^{2+1}}{2+1} + C \\ 2. \int 4 dx = 4x + C \\ 3. \int \frac{dx}{x^2} = -\frac{1}{x} + C \end{array} \right] = \\
 &= \frac{x^3}{3} + 4x + 4 \cdot \left(-\frac{1}{x}\right) + C = \frac{x^3}{3} + 4x - \frac{4}{x} + C
 \end{aligned}$$

$$\begin{aligned}
 8.1.36 \quad \int \frac{dx}{4x^2 + 1} &= \int \frac{dx}{(2x)^2 + 1^2} = \left[\begin{array}{l} 1. \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctg \frac{x}{a} + C \\ 2. t = 2x \Rightarrow dt = d(2x) = (2x)'_x dx = 2 dx \Rightarrow dx = \frac{1}{2} dt \end{array} \right] = \\
 &= \int \frac{\frac{1}{2} dt}{(t)^2 + 1^2} = \frac{1}{2} \int \frac{dt}{t^2 + 1^2} = \frac{1}{2} \cdot \arctg t + C = \\
 &= \frac{1}{2} \arctg(2x) + C
 \end{aligned}$$

$$\begin{aligned}
 8.1.37 \quad \int \left(7^x - \frac{8}{x} + 4 \cos x\right) dx &= \left[\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx \right] = \\
 &= \int 7^x dx - \int \frac{8}{x} dx + \int 4 \cos x dx = \int 7^x dx - 8 \int \frac{dx}{x} + 4 \int \cos x dx = \\
 &= \left[\begin{array}{l} 1. \int a^x dx = \frac{a^x}{\ln a} + C \\ 2. \int \frac{dx}{x} = \ln|x| + C \\ 3. \int \cos x dx = \sin x + C \end{array} \right] = \frac{7^x}{\ln 7} - 8 \ln|x| + 4 \sin x + C
 \end{aligned}$$

$$\begin{aligned}
 8.1.38 \quad \int \left(\frac{\sqrt{3}}{\cos^2 x} - \sqrt[3]{x} - \frac{2}{x^4}\right) dx &= \left[\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx \right] = \\
 &= \int \frac{\sqrt{3}}{\cos^2 x} dx - \int \sqrt[3]{x} dx - \int \frac{2}{x^4} dx = \sqrt{3} \int \frac{dx}{\cos^2 x} - \int x^{\frac{1}{3}} dx - 2 \int x^{-4} dx = \\
 &= \left[\begin{array}{l} 1. \int \frac{dx}{\cos^2 x} = \operatorname{tg} x + C \\ 2. \int x^k dx = \frac{x^{k+1}}{k+1} + C \\ 3. \int x^{-4} dx = \frac{x^{-4+1}}{-4+1} + C \end{array} \right] = \sqrt{3} \cdot \operatorname{tg} x - \frac{x^{\frac{1}{3}+1}}{\frac{1}{3}+1} - 2 \cdot \frac{x^{-4+1}}{-4+1} + C = \\
 &= \sqrt{3} \cdot \operatorname{tg} x + \frac{2}{\sqrt[3]{x}} + \frac{2}{3x^3} + C
 \end{aligned}$$

8.139

$$\begin{aligned}
 \int \left(\sqrt{x} - 3\sqrt[5]{x^3} + 1 \right) dx &= \int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx = \\
 &= \int \sqrt{x} dx - \int \frac{3\sqrt[5]{x^3}}{\sqrt[5]{x}} dx + \int \frac{dx}{\sqrt[5]{x}} = \int x^{\frac{1}{2}} \cdot x^{\frac{1}{5}} dx - 3 \int x^{\frac{3}{5}} \cdot x^{\frac{1}{5}} dx + \int x^{\frac{1}{5}} dx = \\
 &= \int x^{\frac{1}{2} + \frac{1}{5}} dx - 3 \int x^{\frac{3}{5} + \frac{1}{5}} dx + \int x^{\frac{1}{5}} dx = \int x^{\frac{7}{10}} dx - 3 \int x^{\frac{4}{5}} dx + \int x^{\frac{1}{5}} dx = \\
 &= \left[\frac{1}{\frac{7}{10} + 1} \right] x^{\frac{7}{10} + 1} + C = \frac{x^{\frac{17}{10}}}{\frac{17}{10}} - 3 \frac{x^{\frac{9}{5}}}{\frac{9}{5}} + \frac{x^{\frac{6}{5}}}{\frac{6}{5}} + C = \\
 &= \frac{4}{5} x^{\frac{17}{10}} - \frac{60}{25} x^{\frac{9}{5}} + \frac{4}{3} x^{\frac{6}{5}} + C
 \end{aligned}$$

8.140

$$\begin{aligned}
 \int (0,7 \cdot x^{0,9} + 0,2 \cdot (0,5)^x) dx &= \int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx = \\
 &= \int 0,7 \cdot x^{0,9} dx + \int 0,2 \cdot (0,5)^x dx = \left[\int \alpha f(x) dx = \alpha \int f(x) dx \right] = \\
 &= 0,7 \int x^{0,9} dx + 0,2 \int (0,5)^x dx = \left[\begin{array}{l} 1) \int x^a dx = \frac{x^{a+1}}{a+1} + C \\ 2) \int a^x dx = \frac{a^x}{\ln a} + C \end{array} \right] = \\
 &= 0,7 \cdot \frac{x^{-0,1+1}}{-0,1+1} + 0,2 \cdot \frac{(0,5)^x}{\ln 0,5} = \frac{0,7 \cdot x^{0,9}}{0,9} + \frac{0,2 \cdot (0,5)^x}{\ln 0,5}
 \end{aligned}$$

8.141

$$\begin{aligned}
 \int (5 \operatorname{sh} x - 7 \operatorname{ch} x + 1) dx &= \int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx = \\
 &= \int 5 \operatorname{sh} x dx - \int 7 \operatorname{ch} x dx + \int 1 dx = \left[\int \alpha f(x) dx = \alpha \int f(x) dx \right] = \\
 &= 5 \int \operatorname{sh} x dx - 7 \int \operatorname{ch} x dx + \int dx = \left[\begin{array}{l} 1) \int \operatorname{sh} x dx = \operatorname{ch} x + C \\ 2) \int \operatorname{ch} x dx = \operatorname{sh} x + C \\ 3) \int dx = x + C \end{array} \right] = \\
 &= 5 \operatorname{ch} x - 7 \operatorname{sh} x + x + C
 \end{aligned}$$

$$\begin{aligned}
 8.142 \quad \int (x^2-1)(\sqrt{x}+4) dx &= \int (x^2\sqrt{x} + 4x^2 - \sqrt{x} - 4) dx = \\
 &= \int (x^{\frac{5}{2}} + 4x^2 - x^{\frac{1}{2}} - 4) dx = \left[\int f(x) + g(x) dx = \int f(x) dx + \int g(x) dx \right] = \\
 &= \int x^{\frac{5}{2}} dx + \int 4x^2 dx - \int x^{\frac{1}{2}} dx - \int 4 dx = \left[\int df(x) dx = d \int f(x) dx \right] = \\
 &= \int x^{\frac{5}{2}} dx + 4 \int x^2 dx - \int x^{\frac{1}{2}} dx - 4 \int dx = \left[\begin{aligned} (1,2,3) \int x^{\frac{5}{2}} &= \frac{x^{\frac{5}{2}+1}}{\frac{5}{2}+1} + C \\ (4) \int dx &= x \end{aligned} \right] = \\
 &= \frac{x^{\frac{5}{2}+1}}{\frac{5}{2}+1} + 4 \cdot \frac{x^{2+1}}{2+1} - \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} - 4x + C = \\
 &= \frac{2x^{\frac{7}{2}}}{7} + \frac{4x^3}{3} - \frac{2x^{\frac{3}{2}}}{3} - 4x + C = \frac{2x^{\frac{7}{2}}}{7} + \frac{4x^3 - 2x^{\frac{3}{2}}}{3} - 4x + C
 \end{aligned}$$

$$\begin{aligned}
 8.143. \quad \int \frac{7 - \sqrt{x^2+11}}{\sqrt{x^2+11}} dx &= \left[\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx \right] = \\
 &= \int \frac{7}{\sqrt{x^2+11}} dx - \int \frac{\sqrt{x^2+11}}{\sqrt{x^2+11}} dx = \left[\begin{aligned} \frac{d\sqrt{x^2+11}}{dx} &= \frac{x}{\sqrt{x^2+11}}, \text{ т.к. } x^2 > 0 \text{ и } 11 > 0 \\ \Rightarrow \sqrt{x^2+11} &> 0 \end{aligned} \right] = \\
 &= \left[d \int f(x) dx = d \int f(x) dx \right] = 7 \int \frac{dx}{\sqrt{x^2+11}} - \int 1 dx = \\
 &= \left[\begin{aligned} (1) \int \frac{dx}{\sqrt{x^2+11}} &= \ln|x + \sqrt{x^2+11}| + C \\ (2) \int 1 \cdot dx &= x + C \end{aligned} \right] = 7 \ln|x + \sqrt{x^2+11}| - x + C = \\
 &= 7 \ln(x + \sqrt{x^2+11}) - x + C.
 \end{aligned}$$

$$\begin{aligned}
 8.144 \quad \int \left(\frac{\sqrt{x}-5}{x} \right)^3 dx &= \int \frac{(\sqrt{x}-5)^3}{x^3} dx = \int \frac{x\sqrt{x} - 15x + 75\sqrt{x} - 125}{x^3} dx = \\
 &= \int \frac{x^{\frac{3}{2}} - 15x + 75x^{\frac{1}{2}} - 125}{x^3} dx = \int \frac{x^{\frac{3}{2}}}{x^3} - \frac{15x}{x^3} + \frac{75x^{\frac{1}{2}}}{x^3} - \frac{125}{x^3} dx =
 \end{aligned}$$

$$\begin{aligned}
 &= \int \frac{1}{x^{\frac{3}{2}}} - \frac{15}{x^2} + \frac{75}{x^{\frac{5}{2}}} - \frac{125}{x^3} dx = \left[\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx \right] \\
 &= \int \frac{dx}{x^{\frac{3}{2}}} - \int \frac{15}{x^2} dx + \int \frac{75}{x^{\frac{5}{2}}} dx - \int \frac{125}{x^3} dx = \left[\int d(f(x)) = d \int f(x) dx \right] = \\
 &= \int x^{-\frac{3}{2}} dx - 15 \int x^{-2} dx + 75 \int x^{-\frac{5}{2}} dx - 125 \int x^{-3} dx = \left[\frac{1,2}{3,4} \int x^a dx = \frac{x^{a+1}}{a+1} + C \right] = \\
 &= \frac{x^{-\frac{3}{2}+1}}{-\frac{3}{2}+1} - 15 \cdot \frac{x^{-2+1}}{-2+1} + 75 \cdot \frac{x^{-\frac{5}{2}+1}}{-\frac{5}{2}+1} - 125 \cdot \frac{x^{-3+1}}{-3+1} + C = \\
 &= \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}} - 15 \cdot \frac{x^{-1}}{-1} + 75 \cdot \frac{x^{-\frac{3}{2}}}{-\frac{3}{2}} - 125 \cdot \frac{x^{-2}}{-2} + C = \\
 &= -\frac{2}{\sqrt{x}} + \frac{15}{x} - \frac{50}{x\sqrt{x}} + \frac{125}{2x^2} + C
 \end{aligned}$$

8.1.45 $\int \sin 7x dx$ $\left[\begin{aligned} y = 7x &\rightarrow dy = d(7x) = \\ &= (7x)' dx = 7 dx \Rightarrow dx = \frac{1}{7} dy \end{aligned} \right] = \int \sin y \cdot \frac{1}{7} dy =$
 $= \frac{1}{7} \int \sin y dy = \frac{1}{7} \cdot (-\cos y) + C = -\frac{1}{7} \cdot \cos 7x + C$

8.1.46 $\int \sqrt[5]{2x-8} dx = \int (2x-8)^{\frac{1}{5}} dx = \left[\begin{aligned} (1) \int x^a dx &= \frac{x^{a+1}}{a+1} \\ (2) \int \frac{1}{b} f(ax+b) dx &= \frac{1}{a} \cdot F(ax+b) + C \end{aligned} \right] =$
 $= \frac{1}{2} \cdot \frac{(2x-8)^{\frac{1}{5}+1}}{\frac{1}{5}+1} + C = \frac{1}{2} \cdot \frac{5 \cdot 2 \cdot (x-4) \sqrt[5]{2x-8}}{6} + C =$
 $= \frac{5 \cdot (x-4) \cdot \sqrt[5]{2x-8}}{6} + C$

8.1.47 $\int (1-4x)^{2001} dx = \left[\begin{aligned} (1) \int x^a dx &= \frac{x^{a+1}}{a+1} \\ (2) \int f(ax+b) dx &= \frac{1}{a} \cdot F(ax+b) + C \end{aligned} \right] =$
 $= \frac{1}{4} \cdot \frac{(-4x+1)^{2002}}{2002} + C = -\frac{(-4x+1)^{2002}}{8008} + C$

$$8.1.48 \quad \int \frac{dx}{9x+7} = \left[\begin{array}{l} (1) \int \frac{dx}{x} = \ln|x| + C \\ (2) \int f(ax+b) dx = \frac{1}{a} \cdot F(ax+b) + C \end{array} \right] =$$

$$= \frac{1}{9} \cdot \ln|9x+7| + C = \underline{\underline{\frac{\ln|9x+7|}{9} + C}}$$

$$8.1.49 \quad \int \frac{dx}{(6x+11)^4} = \int (6x+11)^{-4} dx = \left[\begin{array}{l} (1) \int x^a dx = \frac{x^{a+1}}{a+1} + C \\ (2) \int f(ax+b) dx = \frac{1}{a} \cdot F(ax+b) + C \end{array} \right] =$$

$$= \frac{1}{6} \cdot \frac{(6x+11)^{-3}}{-3} + C = \underline{\underline{-\frac{1}{18 \cdot (6x+11)^3} + C}}$$

$$8.1.50 \quad \int \frac{dx}{25x^2+1} = \int \frac{dx}{(5x)^2+1} = \left[\begin{array}{l} y=5x \Rightarrow dy=d(5x)= \\ (5x)'_x dx = 5dx \Rightarrow dx = \frac{1}{5} dy \end{array} \right] =$$

$$= \int \frac{\frac{1}{5} dy}{y^2+1} = \frac{1}{5} \int \frac{dy}{y^2+1} = \frac{1}{5} \cdot \arctan y + C = \underline{\underline{\frac{\arctan(5x)}{5} + C}}$$

$$8.1.51 \quad \int 3^{2-11x} dx = \int 3^{-11x+2} dx = \left[\begin{array}{l} (1) \int a^x dx = \frac{a^x}{\ln a} + C \\ (2) \int f(ax+b) dx = \frac{1}{a} \cdot F(ax+b) + C \end{array} \right] =$$

$$= -\frac{1}{11} \cdot \frac{3^{-11x+2}}{\ln 3} + C = \underline{\underline{-\frac{3^{-11x+2}}{11 \cdot \ln 3} + C}}$$

$$8.1.52 \quad \int \frac{dx}{\sqrt{4x^2-1}} = \int \frac{dx}{\sqrt{(2x)^2-1}} = \left[\begin{array}{l} y=2x \Rightarrow dy=d(2x)= \\ (2x)'_x dx = 2dx \Rightarrow dx = \frac{1}{2} dy \end{array} \right] =$$

$$= \int \frac{\frac{1}{2} dy}{\sqrt{y^2-1}} = \frac{1}{2} \int \frac{dy}{\sqrt{y^2-1}} = \frac{1}{2} \cdot \ln|y+\sqrt{y^2-1}| + C =$$

$$= \underline{\underline{\frac{1}{2} \cdot \ln|2x+\sqrt{4x^2-1}| + C}}$$

8.1.58

$$\begin{aligned}
 \int \frac{4\sqrt{1-x^2} + 3x^2}{x^2-1} dx &= \int \frac{4\sqrt{1-x^2}}{x^2-1} dx + \int \frac{3x^2}{x^2-1} dx = \\
 &= -4 \int \frac{\sqrt{1-x^2}}{1-x^2} dx + 3 \int \frac{x^2}{x^2-1} dx = -4 \int \frac{dx}{\sqrt{1-x^2}} + 3 \int \frac{x^2-1+1}{x^2-1} dx = \\
 &= -4 \int \frac{dx}{\sqrt{1-x^2}} + 3 \int \left(\frac{x^2-1}{x^2-1} + \frac{1}{x^2-1} \right) dx = -4 \int \frac{dx}{\sqrt{1-x^2}} + 3 \int 1 dx + 3 \int \frac{dx}{x^2-1} = \\
 &= \left[\begin{array}{l} 1) \int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin \frac{x}{a} + C \\ 2) \int 1 dx = x + C \\ 3) \int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C \end{array} \right] = -4 \arcsin x + 3x + \frac{3}{2} \ln \left| \frac{x-1}{x+1} \right| + C
 \end{aligned}$$

8.1.59.

$$\begin{aligned}
 \int \frac{\cos 2x dx}{\sin^2 x \cos^2 x} &= \left[\cos 2x = \cos^2 x - \sin^2 x \right] = \int \frac{\cos^2 x - \sin^2 x}{\sin^2 x \cos^2 x} dx = \\
 &= \int \frac{\cos^2 x}{\sin^2 x \cos^2 x} dx - \int \frac{\sin^2 x}{\sin^2 x \cos^2 x} dx = \int \frac{dx}{\sin^2 x} - \int \frac{dx}{\cos^2 x} = \\
 &= \left[\begin{array}{l} 1) \int \frac{dx}{\sin^2 x} = -\cot x + C \\ 2) \int \frac{dx}{\cos^2 x} = \tan x + C \end{array} \right] = -\cot x - \tan x + C
 \end{aligned}$$

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8.1.60

$$\begin{aligned}
 \int \frac{\sin 2x}{\cos x} dx &= \left[\sin 2x = 2 \sin x \cos x \right] = \int \frac{2 \sin x \cos x}{\cos x} dx = 2 \int \sin x dx = \\
 &= -2 \cos x + C
 \end{aligned}$$